

WEEK 01

Prompt Engineering & ReAct

Logic Puzzle:

A farmer needs to transport a wolf, a goat, and a cabbage across a river.

The boat can carry only the farmer and one item at a time.

Rules:

- The wolf cannot be left alone with the goat.
- The goat cannot be left alone with the cabbage.

Find the correct sequence of moves to get all three across the river safely.

CoT Prompt:

You are an expert reasoning assistant.

Solve the following river crossing puzzle.

Think step by step and explain every decision before giving the final answer.

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Find the correct sequence of moves.

GPT RESPONSE OF CoT Prompt:

ChatGPT ▾

✓ **Correct final sequence**

1. Take Goat → Right
 - Left: Wolf, Cabbage | Right: Goat
2. Return alone
 - Left: Wolf, Cabbage, Farmer | Right: Goat
3. Take Wolf → Right
 - Left: Cabbage | Right: Wolf, Goat
4. Bring Goat → Left
 - Left: Goat, Cabbage | Right: Wolf
5. Take Cabbage → Right
 - Left: Goat | Right: Wolf, Cabbage
6. Return alone
 - Left: Goat | Right: Wolf, Cabbage
7. Take Goat → Right
 - Left: — | Right: Wolf, Goat, Cabbage



+ Ask anything

ReAct Prompt:

You are an AI agent that uses the ReAct framework.

For every step use:

Thought:

Action:

Observation:

Then provide:

Final Answer:

Puzzle:

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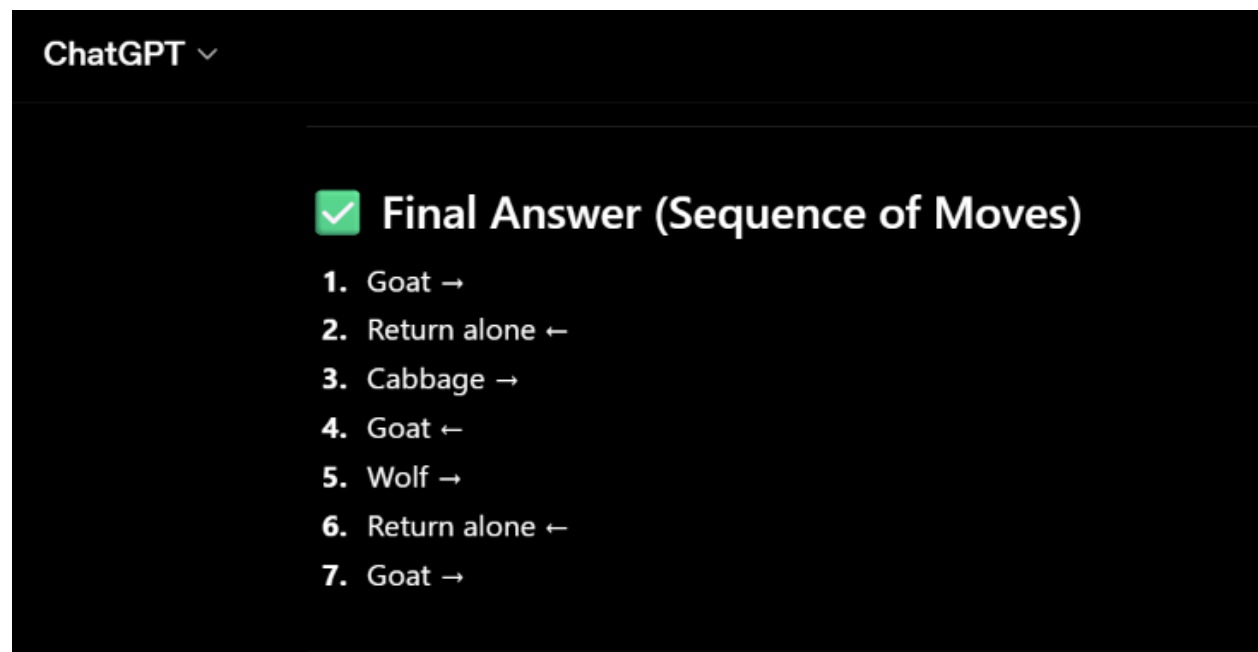
The boat can carry only the farmer and one item at a time.

Rules:

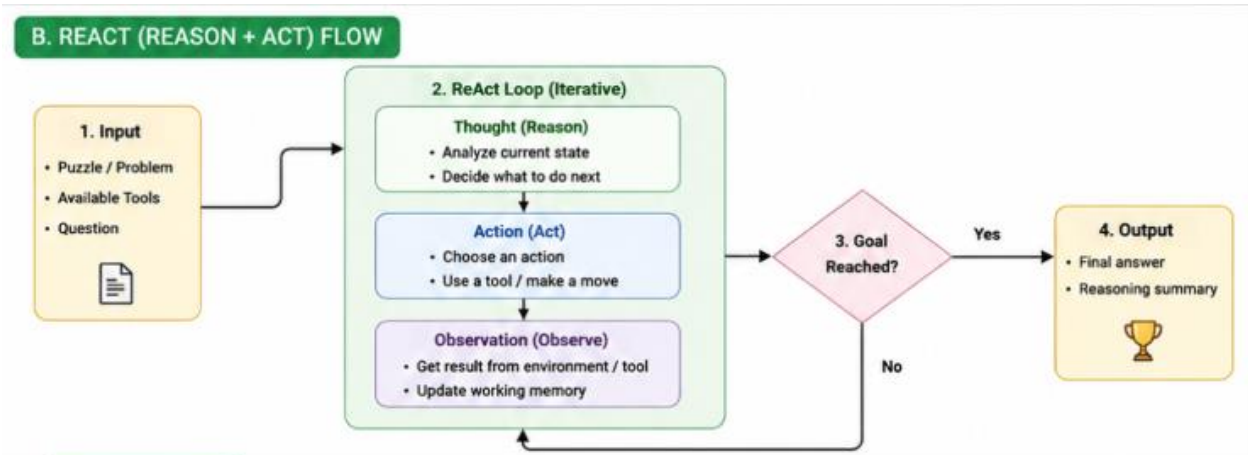
- The wolf cannot be left alone with the goat.
- The goat cannot be left alone with the cabbage.

Find the correct sequence of moves.

GPT RESPONSE OF **ReAct Prompt**:



LOGIC FLOW DIAGRAM OF ReAct



LOGIC FLOW DIAGRAM OF CoT

